Textual Prior Enhancement and Spatially Guided Context Fusion for Human-Object Interaction Detection

Ke Wang[a,*,1], Yuhao Jiang[b], Chuang Qiu[c] and Weilin Gao[d]

[a]Chongqing University, , Chongqing, 400030, China

ARTICLE INFO

Keywords:
Human-object interaction detection
Vision-language models
Textual priors
Semantic denoising
Spatially guided context fusion

ABSTRACT

Human-object interaction (HOI) detection aims to localize humans and objects while recognizing the interactions between them. Although vision-language models (VLMs) offer rich semantic priors for HOI detection, their direct integration into dense prediction frameworks may weaken fine-grained visual representations and introduce irrelevant background semantics. Moreover, visually similar interactions and the long-tailed distribution of HOI categories remain challenging. To address these issues, we propose the Semantic Prior-guided Adaptive Fusion Network (SPAF-Net), an end-to-end framework for HOI detection. First, a text-enhanced dual-stream encoder injects offline textual priors into visual memory via gated residual cross-modal fusion, effectively leveraging semantic knowledge while preserving discriminative visual cues. Second, we introduce a semantic denoising strategy that perturbs ground-truth labels according to text-derived semantic similarities, thereby constructing long-tail-aware hard negative samples and improving the detector's ability to distinguish visually confusing actions. Third, a spatially guided adaptive context fusion module uses progressively predicted human and object regions to extract relevant VLM image features, suppress background interference, and adaptively integrate the fused contextual information into the human, object, and interaction branches. Extensive experiments on the HICO-DET and V-COCO benchmarks demonstrate that SPAF-Net achieves significant advantages, improving the Full mAP under the default setting by 0.41% and 4.41% over PVIC and SCTC++, respectively. The code is publicly avaliable at https://github.com/JYH527/SPAF-Net.git

1. Introduction

HOI detection is defined as a problem of the ⟨human, object, interaction⟩ triplet detection Guo, Lan, Yang, Liu and Gao (2022). It detects not only the categories and positions of humans and objects but also the interactions between them Zhang, Yuan, Campbell, Zhong and Gould (2023). Most of the existing research methods only focus on the accurate location and classification of humans and objects Zhang, Li, Liu, Zhang, Su, Zhu, Ni and Shum (2022b), Zhao, Lv, Xu, Wei, Wang, Dang, Liu and Chen (2024), but fail to deeply understand the interaction between them. HOI detection can make up for this deficiency Zou, Wang, Hu, Liu, Wu, Zhao, Li, Zhang, Zhang, Wei et al. (2021), and enable the machine to truly understand the dynamic interaction between people and objects like humans, rather than being limited to static descriptions of scene appearance Kim, Lee, Kang, Kim and Kim (2021). Since HOI detection solves the problem of human-centered object interaction, it is a fundamental step for behavior understanding and can be further applied to multiple downstream domains.

Traditional HOI detection methods, whether adopting a two-stage pipeline Ulutan, Iftekhar and Manjunath (2020), Hou, Peng, Qiao and Tao (2020) or a one-stage parallel paradigm Tamura, Ohashi and Yoshinaga (2021), Kim et al. (2021), predominantly rely on isolated visual representations. While these visual-only models excel at localizing entities, they severely struggle with the combinatorial explosion of human-object pairs and the long-tail distribution inherent in HOI datasets. More critically, purely visual paradigms suffer from fundamental "visual ambiguity". Fine-grained interactions (e.g., "hold" versus "carry") often exhibit nearly identical visual poses and human-object relative position. Without high-level linguistic constraints, visual-only models lack deep semantic comprehension, causing them to overfit to superficial spatial relationship rather than learning genuine interactive semantics.

To mitigate these representational bottlenecks, recent works have integrated Vision-Language Models (VLMs) and textual priors into HOI detection Yuan, Jiang, Albanie, Feng, Huang, Ni and Tang (2022a), Ning, Qiu, Liu and He (2023), Liao, Zhang, Lu, Wang, Li and Liu (2022). By mapping visual regions into a structured semantic space, VLMs provide powerful relational priors that alleviate long-tail issues and enhance fine-grained recognition. However, directly migrating VLMs to dense interaction prediction introduces new challenges. One primary obstacle is that unconstrained cross-modal fusion often leads to visual feature suppression, where dominant language priors overwhelm the fine-grained local visual cues. Furthermore, the extraction of global VLM features inevitably incorporates irrelevant background clutter, which severely contaminates highly localized interactive regions and consequently disrupts precise interaction reasoning.

To systematically address these challenges, we propose a novel end-to-end multi-modal framework for HOI detection, as shown in Fig. 1. To prevent linguistic priors from

*Corresponding author at: College of Mechanical and Vehicle Engineering, Chongqing University, Chongqing 400030, China.

kw@cqu.edu.cn (K. Wang); 202407131487@stu.cqu.edu.cn (Y. Jiang); 20213971@stu.cqu.edu.cn (C. Qiu); weilingao@stu.cqu.edu.cn (W. Gao)

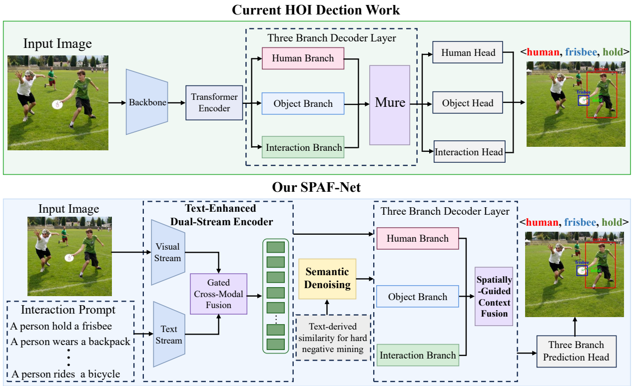

Figure 1: Comparison between current work and our proposed SPAF-Net

overshadowing intrinsic fine-grained visual features, we introduce a residual gated fusion mechanism within the dual-stream encoder, enabling the safe incorporation of textual knowledge into visual representations. To tackle the long-tail distribution and fine-grained recognition challenges, we devise a novel semantic denoising strategy. This strategy probabilistically perturbs ground-truth labels toward semantically similar alternatives during training, thereby synthesizing hard negative samples that compel the model to capture discriminative fine-grained action cues. Furthermore, a scaling factor is applied to the perturbation probability of tail categories, deliberately exposing the model to intensified long-tail denoising scenarios during training, which effectively mitigates the severe class imbalance inherent in HOI datasets. The proposed spatially-guided Adaptive Context Fusion (SG-ACF) module disentangles interaction semantics from background clutter by using predicted bounding boxes as spatial masks to precisely retrieve region-specific features from VLMs. In summary, the primary contributions of this work are as follows:

- **Text-Enhanced Dual-Stream Encoder with Gated Fusion (TDE-GF):** We design a dual-stream encoder to supplement visual features with pre-extracted textual priors, employing a residual gated fusion mechanism to prevent textual semantics from overshadowing intrinsic fine-grained visual features.
- **Semantic Denoising for Hard Negative Mining (SD-HNM):** By flipping ground-truth labels to semantically similar ones and applying a tail-category scaling factor, we construct explicit hard negative samples which compels the network to capture fine-grained actions and mitigates long-tail degradation.
- **Spatially-Guided Adaptive Context Fusion (SG-ACF):** We utilize the predicted bounding boxes at each layer as spatial masks. This enables the precise retrieval of region-specific VLM features, fundamentally preventing global noise from contaminating local interactions.
- **State-of-the-Art Performance:** Extensive experiments on HICO-DET Chao, Liu, Liu, Zeng and Deng (2018) and V-COCO Gupta and Malik (2015) show that our framework outperforms strong methods, including SCTC++ Ren, Luo, Jiang, Qu, Han, Tian and Liu (2024) and PVIC Zhang et al. (2023).

2. Related Work

2.1. VM-based HOI Detection

Conventionally, HOI detection methods are broadly categorized into two-stage and one-stage paradigms. Two-stage approaches Ulutan et al. (2020) decouple entity localization from interaction prediction, frequently employing Graph Convolutional Networks (GCNs) Wang, Jiao, Liu, Li, Liu, Ji and Gan (2021); Su, Wang, Xie and Yu (2022); Park, Park and Lee (2023) or multi-stream architectures Hou, Yu, Qiao, Peng and Tao (2021); Zhang et al. (2023) to model structural topologies and integrate auxiliary spatial cues. Conversely, one-stage methods formulate HOI detection as a unified, parallel prediction task Tamura et al. (2021); Junwen and Keiji (2023). Spearheaded by the paradigm shift of QPIC Tamura et al. (2021), which leveraged the DETR framework to reformulate the task as direct triplet prediction, subsequent research has extensively optimized this architecture. Notable innovations include cascaded decoders for decoupled localization and classification Zhang, Liao, Liu, Lu, Wang, Gao and Li (2021), dual-branch transformers for dedicated triplet modeling Guo et al. (2022); Kim et al. (2021), and multi-scale deformable attention mechanisms aimed at generating fine-grained interaction proposals Junwen and Keiji (2023); Ma, Wang, Wang and Wei (2023). Despite these architectural strides, purely vision-based models inherently rely on isolated visual representations. They fundamentally lack the high-level semantic constraints required to resolve visual ambiguity.

2.2. VLM-based HOI Detection

Recent advancements leverage the rich contextual features of pre-trained Vision-Language Models (VLMs) Radford, Kim, Hallacy, Ramesh, Goh, Agarwal, Sastry, Askell, Mishkin, Clark et al. (2021) to address the representational bottlenecks of visual-only networks. Methods like GEN-VLKT Liao et al. (2022) and HOICLIP Ning et al. (2023) transfer visual-linguistic knowledge from CLIP by initializing classifiers with text embeddings or utilizing cross-attention interaction decoders, significantly improving few-shot and zero-shot generalization. To mitigate long-tailed distributions, ADA-CM Lei, Caba, Chen, Jin, Peng and Liu (2023) employs a concept-guided memory alongside an instance-aware adapter mechanism. Furthermore, approaches such as SOV-STG Chen, Wang and Yanai (2025) integrate vision-language advisors and multi-modal spatial priors to enrich complex interaction reasoning and reduce misclassifications. However, directly migrating VLM priors into dense HOI prediction introduces new challenges: unconstrained cross-modal fusion often suppresses fine-grained local visual cues, and the extraction of global semantic features inevitably incorporates background clutter that contaminates highly localized interactive regions.

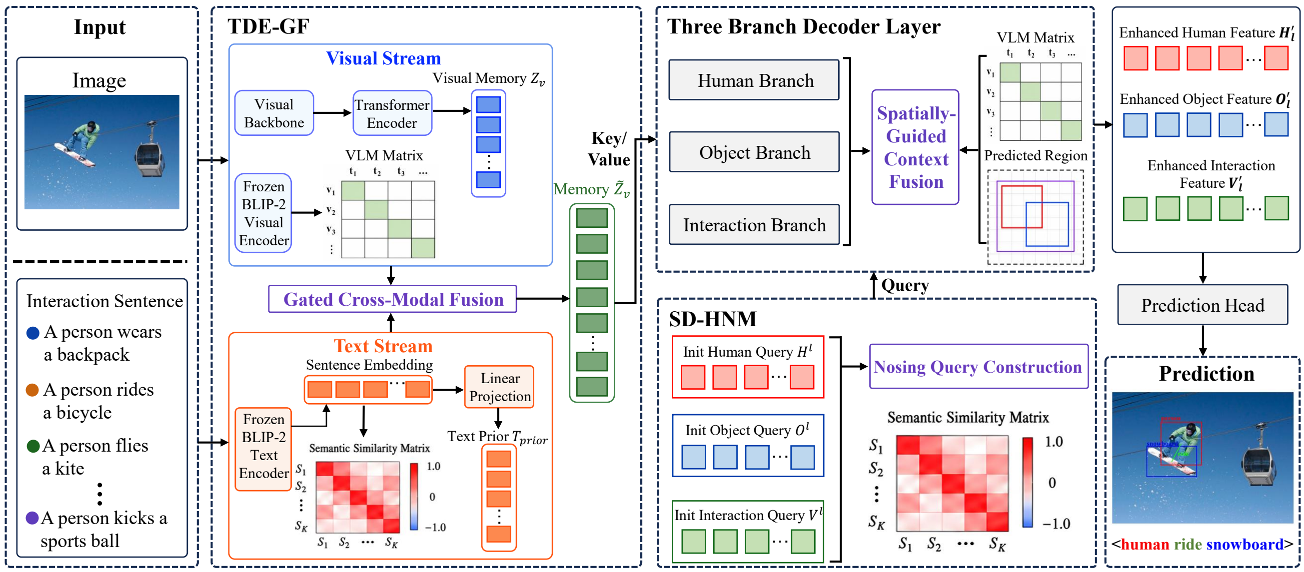

Figure 2: Overview of the proposed SPAF-Net framework. The text-enhanced dual-stream encoder injects semantic priors into visual memory through gated residual fusion. Semantic denoising constructs text-informed hard negatives, while the three-branched decoder and SG-ACF jointly predict human boxes, object boxes, object categories, and interaction labels.

3. PROPOSED METHOD

3.1. Method Overview

Given an image I, the objective of HOI detection is to predict a set of interaction triplets

$$\mathcal{Y} = \{(\hat{b}_h, \hat{b}_o, \hat{c}_o, \hat{\mathbf{y}}_v)\}_{i=1}^{N_q}, \tag{1}$$

where $\hat{b}_h$ and $\hat{b}_o$ denote the human and object bounding boxes, $\hat{c}_o$ is the object category, $\hat{\mathbf{y}}_v \in [0,1]^K$ is the multi-label verb-probability vector over K interaction categories, and N_q is the number of object queries. As shown in Fig. 2, our framework follows a DETR-style set prediction paradigm while explicitly incorporating textual priors and spatially localized vision-language context. The overall architecture consists of four key components: (1) a text-enhanced dual-stream encoder that injects semantic priors into visual memory through gated cross-attention; (2) a semantic denoising strategy that constructs long-tail hard negatives from the textual similarity structure; (3) a three-branched decoder that disentangles human, object, and relation representations; and (4) a spatially-guided adaptive context fusion module that constructs interaction-aware contextual representations by aggregating VLM patch features within predicted human-object regions.

3.2. Text-Enhanced Dual-Stream Encoder with Gated Fusion

Purely visual HOI detectors are vulnerable to visual ambiguity, since interactions such as *hold*, *carry*, and *lift* may exhibit highly similar poses and spatial layouts. Although language priors provide high-level relational knowledge, unconstrained fusion can suppress local visual evidence that is essential for accurate localization and fine-grained action recognition. We therefore design a text-enhanced dual-stream encoder that preserves discriminative visual representations while using textual semantics as a controlled structural prior. The encoder structure is shown in Fig. 3.

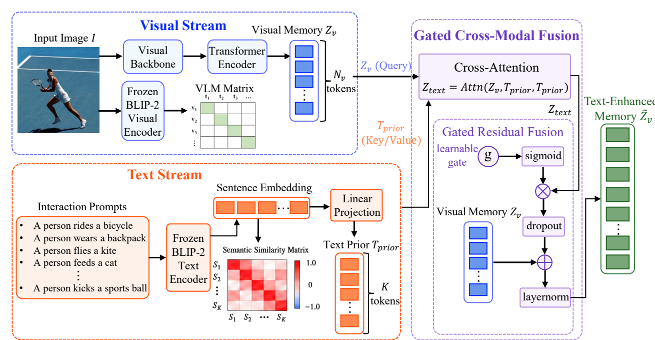

Figure 3: Text-enhanced dual-stream encoder with gated fusion. Visual memory queries the text-prior bank through cross-attention, and a learnable residual gate controls the injected semantics to preserve discriminative visual evidence.

3.2.1. *Visual Stream*

Given an input image $I \in \mathbb{R}^{B\times 3\times H_0\times W_0}$, where B is the batch size and H_0, W_0 are the input height and width, a convolutional backbone followed by a Transformer encoder $\mathcal{E}_{vis}$ extracts visual memory tokens

$$Z_v = \mathcal{E}_{vis}(I) \in \mathbb{R}^{B\times N_v\times D}, \tag{2}$$

where $N_v = H \times W$ denotes the flattened spatial length after backbone downsampling, and D is the detector hidden dimension. Two-dimensional sine positional encodings are added to retain spatial ordering. In parallel, a frozen BLIP-2 visual encoder $\mathcal{E}_{vlm}^{img}$ extracts vision-language-aligned patch tokens

$$V_{vlm} = \mathcal{E}_{vlm}^{img}(I) \in \mathbb{R}^{B\times N_p\times D_{vlm}}, \tag{3}$$

where N_p is the number of VLM patch tokens and D_{vlm} is the VLM embedding dimension. A learned linear projection maps V_{vlm} from D_{vlm} to the detector dimension D when it is used by the subsequent context fusion module.

3.2.2. *Text Stream*

To construct an offline semantic prior bank, we describe each predefined interaction category with a natural language prompt:

$$\mathcal{S} = \{S_1, S_2, \ldots, S_K\}. \tag{4}$$

where K is the number of predefined interaction categories and S_i is the prompt for category i. The frozen BLIP-2 text encoder maps each sentence S_i into token embeddings $\{E_{i,j}\}_{j=1}^{L_i}$, where L_i is the token count and $E_{i,j} \in \mathbb{R}^{D_{vlm}}$ is the embedding of token j. To obtain a robust sentence-level representation, we apply masked mean pooling:

$$T_i = \frac{\sum_{j=1}^{L_i} E_{i,j} \mathcal{M}_{i,j}}{\max\left(\sum_{j=1}^{L_i} \mathcal{M}_{i,j}, \epsilon\right)} \in \mathbb{R}^{D_{vlm}}, \tag{5}$$

where $\mathcal{M}_{i,j} \in \{0,1\}$ indicates whether the j-th token is valid and ϵ avoids division by zero. The resulting text embeddings are projected into the detector dimension by a learned projection ϕ_t to form

$$T_{prior} = \phi_t([T_1; T_2; \ldots; T_K]) \in \mathbb{R}^{K \times D}. \tag{6}$$

where $[T_1; \ldots; T_K]$ stacks the K category embeddings along the category axis. We further compute a semantic similarity matrix from the normalized text embeddings:

$$S_{i,j} = \frac{T_i^\top T_j}{\|T_i\|_2 \|T_j\|_2}, \quad i, j \in \{1, \ldots, K\}. \tag{7}$$

This matrix encodes the semantic proximity among interaction categories and is later used to sample informative denoising labels; its entries lie in $[-1, 1]$ because they are cosine similarities.

3.2.3. *Gated Cross-Modal Fusion*

To incorporate textual priors without overwhelming visual evidence, we use the visual memory as queries and the text prior bank as keys and values:

$$C_{text} = \mathrm{Softmax}\left(\frac{(Z_v W_Q)(T_{prior} W_K)^\top}{\sqrt{D}}\right)(T_{prior} W_V), \tag{8}$$

where W_Q, W_K, and W_V are learnable query, key, and value projections with compatible output dimension D. The context $C_{text} \in \mathbb{R}^{B \times N_v \times D}$, and the softmax is applied over the K text-prior entries for each visual token. The final text-enhanced memory is obtained through a gated residual update:

$$\tilde{Z}_v = \mathrm{LN}\left(Z_v + \mathrm{Dropout}\left(\sigma(g) \odot C_{text}\right)\right), \tag{9}$$

where $\mathrm{LN}(\cdot)$ denotes layer normalization, $\sigma(\cdot)$ is the sigmoid function, $\odot$ denotes element-wise multiplication, and g is a learnable scalar or channel-wise gate broadcast over tokens. Initializing g close to zero stabilizes early training and lets the detector progressively determine how much linguistic context should be introduced.

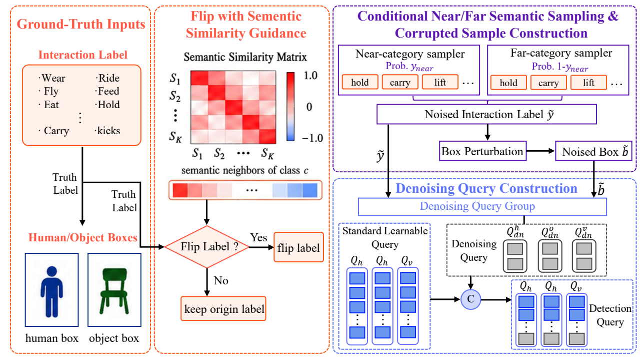

Figure 4: Semantic denoising for hard-negative mining. During training, ground-truth labels are perturbed according to text-derived semantic similarity, with a tail-aware schedule that increases supervision for infrequent interaction categories.

3.3. Semantic Denoising for Hard Negative Mining

DETR-based HOI detectors rely on bipartite matching, which provides sparse supervision and may slow convergence. Denoising training alleviates this issue by feeding perturbed ground-truth annotations into the decoder. However, uniform label corruption does not reflect the fine-grained semantic confusion and long-tail distribution of HOI datasets. We thus introduce Semantic Denoising (SemDN), which uses the textual similarity matrix in Eq. (7) to synthesize hard negative interaction labels. The semantic perturbation and query construction process is summarized in Fig. 4.

During training, we maintain three groups of fixed learnable queries for the standard detection branch:

$$Q_h, Q_o, Q_r \in \mathbb{R}^{N_q \times D}, \tag{10}$$

corresponding to the human, object, and relation streams. For the denoising branch, ground-truth boxes are perturbed by random center shifts and scale changes. For a positive verb class $c \in \{1, \ldots, K\}$, the semantic label corruption probability at epoch t is defined as

$$p_{flip}^{(c)}(t) = \mathrm{clip}\left(\left(p_{\min} + \Delta p \cdot \rho(t)\right)\left[1 + \beta(1 - f_c)\right], 0, 1\right), \tag{11}$$

where $p_{\min}$ is the initial corruption probability, Δp is the scheduled increment, $\rho(t) \in [0,1]$ is the warm-up ratio, $f_c \in [0,1]$ is the normalized category frequency, and β controls the additional emphasis on tail categories. The operator $\mathrm{clip}(\cdot, 0, 1)$ limits the resulting probability to $[0, 1]$.

If corruption is activated, a replacement label $\tilde{y}$ is sampled from semantic neighborhoods induced by the c-th row of S:

$$\tilde{y} \sim \begin{cases} \mathrm{Categorical}(\mathcal{P}_{near}^{(c)}), & \text{with probability } \gamma_{near}, \\ \mathrm{Categorical}(\mathcal{P}_{far}^{(c)}), & \text{with probability } 1 - \gamma_{near}, \end{cases} \tag{12}$$

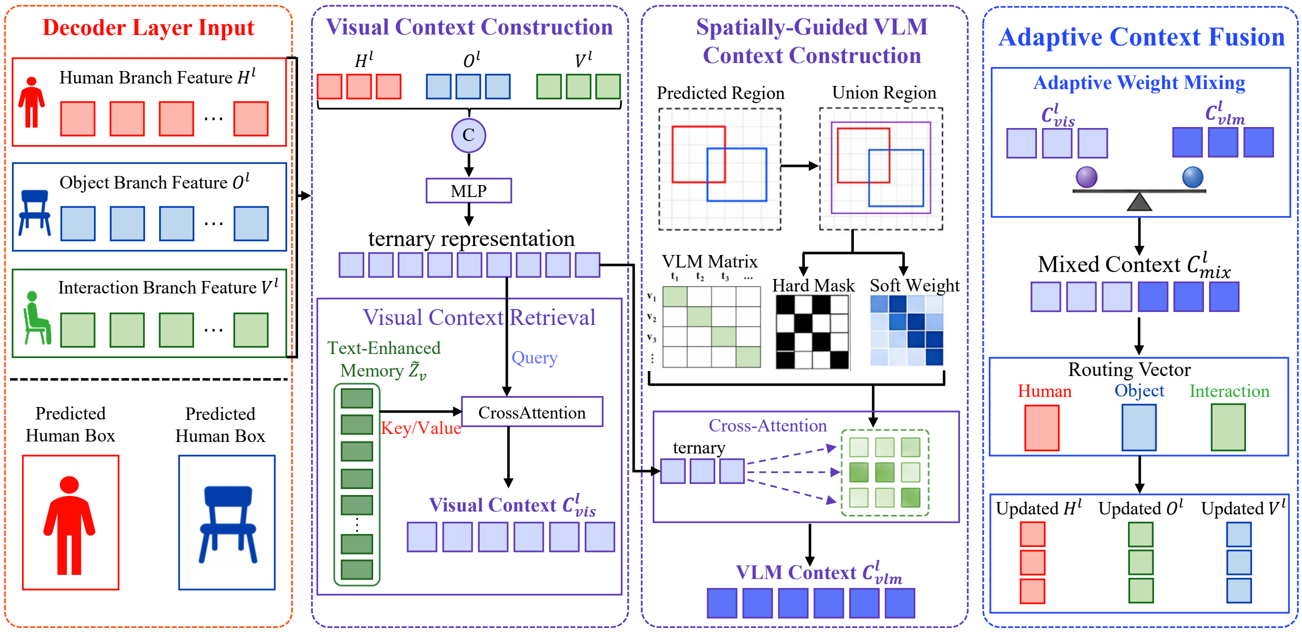

Figure 5: Spatially-Guided Adaptive Context Fusion (SG-ACF). Predicted human and object boxes define spatial masks over VLM patch tokens; the retrieved context is adaptively mixed with detector features and routed to the human, object, and interaction branches.

where $\mathcal{P}_{near}^{(c)}$ and $\mathcal{P}_{far}^{(c)}$ are normalized categorical distributions derived from the c-th row of S, $\gamma_{near} \in [0, 1]$ is the probability of selecting the near-category branch, and $\tilde{y}$ is the sampled replacement verb label. The near-category branch encourages the model to distinguish visually similar interactions, while the far-category branch preserves robustness to broader label perturbations.

$$Q_k^{train} = [Q_k; Q_k^{dn}], \quad k \in \{h, o, r\}. \tag{13}$$

where $Q_k^{dn} \in \mathbb{R}^{N_{dn} \times D}$ denotes the denoising queries constructed from the perturbed boxes and labels. The noised labels and boxes are embedded by lightweight MLPs to form denoising queries, which are concatenated with the standard learnable queries Q_k, during training. Only the standard queries are used during inference.

3.4. Three-Branched Decoder

Human localization, object recognition, and interaction classification require different visual cues. A shared decoder can entangle these heterogeneous objectives, especially when the relation branch must focus on contextual and geometric evidence rather than object appearance alone. We therefore adopt a three-branched Transformer decoder, where the human, object, and relation branches are decoded in parallel from the text-enhanced memory:

$$H^\ell, O^\ell, R^\ell = \mathcal{D}^\ell(Q_h^\ell, Q_o^\ell, Q_r^\ell, \tilde{Z}_v), \quad \ell = 1, \dots, L_d, \tag{14}$$

where $\mathcal{D}^\ell$ denotes decoder layer ℓ, $Q_h^\ell, Q_o^\ell, Q_r^\ell$ are the corresponding layer inputs, L_d is the number of decoder layers, and $H^\ell, O^\ell, R^\ell \in \mathbb{R}^{B \times N_q \times D}$ denote the decoded human, object, and relation features. Each branch contains self-attention for query interaction and cross-attention for memory retrieval. The decoded representations are used to predict human boxes, object boxes, object categories, and verb scores, and also provide layer-wise inputs for spatially guided context fusion.

3.5. Spatially-Guided Adaptive Context Fusion

Directly attending to all VLM patch tokens introduces background semantics that are not necessarily related to the current human-object pair. To obtain interaction-specific multi-modal evidence, we propose Spatially-Guided Adaptive Context Fusion (SG-ACF), which uses predicted boxes to constrain VLM context retrieval. An overview of SG-ACF is provided in Fig. 5.

At decoder layer ℓ, the triplet representation is first formed by aggregating the three branch features:

$$Q_{tri}^\ell = \phi_{tri}\left([H^\ell; O^\ell; R^\ell]\right) \in \mathbb{R}^{B \times N_q \times D}, \tag{15}$$

where $\phi_{tri}(\cdot)$ is an MLP and $[\cdot;\cdot]$ denotes channel-wise concatenation. The operator $\mathrm{Attn}(Q, K, V)$ denotes scaled dot-product attention with query, key, and value inputs; it produces the visual context vector

$$C_{vis}^\ell = \mathrm{Attn}(Q_{tri}^\ell, \tilde{Z}_v, \tilde{Z}_v) \in \mathbb{R}^{B \times N_q \times D}. \tag{16}$$

To localize the relevant VLM tokens, we compute the union region of the current human and object boxes and adaptively enlarge it with a learned offset:

$$b_u^\ell = \hat{b}_h^\ell \cup \hat{b}_o^\ell, \tag{17}$$

$$b_a^\ell = \mathrm{clip}\left(b_u^\ell + \tanh(W_\Delta Q_{tri}^\ell)\lambda_s, 0, 1\right). \tag{18}$$

where $\hat{b}_h^\ell$ and $\hat{b}_o^\ell$ are the normalized human and object boxes predicted at layer ℓ, b_u^ℓ is their coordinate-wise union, b_a^ℓ denotes the adaptive spatial margin bounding region, W_Δ predicts the box offset, and λ_s controls the maximum expansion range. The operator clip$(\cdot, 0, 1)$ constrains normalized box coordinates to $[0, 1]$. The box coordinates used to construct the region mask are detached from the computational graph so that context retrieval does not destabilize box regression.

For each VLM patch location, we build a binary hard mask $M^{\ell}_{hard} \in \{0,1\}^{B\times N_q\times N_p}$ and a continuous soft weight $W^{\ell}_{soft} \in (0,1]^{B\times N_q\times N_p}$ according to its spatial relation to b^{ℓ}_{a}. The spatially biased VLM attention is then computed as

$$A^{\ell}_{vlm} = \mathrm{Softmax}\left(\Omega^{\ell}\right),$$
$$\Omega^{\ell} = \frac{(Q^{\ell}_{tri}W^{vlm}_{Q})(V_{vlm}W^{vlm}_{K})^{\top}}{\sqrt{d_k}} + \log W^{\ell}_{soft} + \log M^{\ell}_{hard}, \tag{19}$$

where W^{vlm}_{Q}, W^{vlm}_{K}, and W^{vlm}_{V} are VLM-attention query, key, and value projections, d_k is the key dimension, and $\log M^{\ell}_{hard}$ is implemented as 0 for valid patches and $-\infty$ for invalid patches. Both A^{ℓ}_{vlm} and Ω^{ℓ} have shape $B \times N_q \times N_p$, and the softmax is applied over the N_p patch tokens. This formulation applies both hard spatial exclusion and soft spatial bias before normalization, yielding a properly normalized attention distribution. The VLM context is

$$C^{\ell}_{vlm} = A^{\ell}_{vlm}(V_{vlm}W^{vlm}_{V}) \in \mathbb{R}^{B\times N_q\times D}. \tag{20}$$

The visual and VLM contexts are adaptively mixed, The mixing process is as follows:

$$\alpha^{\ell} = \sigma\left(\phi_{\alpha}([Q^{\ell}_{tri}; C^{\ell}_{vis}; C^{\ell}_{vlm}])\right) \in [0,1]^{B\times N_q\times D}, \tag{21}$$

$$C^{\ell}_{mix} = \alpha^{\ell} \odot C^{\ell}_{vlm} + (1-\alpha^{\ell}) \odot C^{\ell}_{vis} \in \mathbb{R}^{B\times N_q\times D}. \tag{22}$$

where ϕ_{α} is a learned gating MLP, σ maps the gate to $[0, 1]$, and α^{ℓ} is broadcast over feature channels. Finally, a routing MLP ϕ_r produces a three-way routing vector that distributes the mixed context back to the decoder branches:

$$\mathbf{w}^{\ell} = \mathrm{Softmax}\left(\phi_r([H^{\ell}; O^{\ell}; R^{\ell}; C^{\ell}_{mix}])\right) \in \mathbb{R}^{B\times N_q\times 3}, \tag{23}$$

$$\begin{aligned} \bar{H}^{\ell} &= H^{\ell} + w^{\ell}_{h}C^{\ell}_{mix}, \\ \bar{O}^{\ell} &= O^{\ell} + w^{\ell}_{o}C^{\ell}_{mix}, \\ \bar{R}^{\ell} &= R^{\ell} + w^{\ell}_{r}C^{\ell}_{mix}. \end{aligned} \tag{24}$$

where $w^{\ell}_{h}, w^{\ell}_{o}, w^{\ell}_{r}$ are the three per-query entries of $\mathbf{w}^{\ell}$ and sum to one after the softmax. The routed features are passed to the prediction heads, enabling the detector to exploit local VLM semantics while suppressing irrelevant background context.

3.6. Loss Function

The model is trained end-to-end with a set prediction objective. Hungarian matching assigns each ground-truth HOI triplet to one prediction by considering box regression, object classification, and verb classification costs. After matching, the overall loss is

$$\begin{aligned} \mathcal{L}_{total} = \lambda_{obj}\mathcal{L}_{obj} + \lambda_{verb}\mathcal{L}_{verb} + \lambda_{L1}\mathcal{L}_{L1} \\ + \lambda_{giou}\mathcal{L}_{giou} + \lambda_{dn}\mathcal{L}_{dn}, \end{aligned} \tag{25}$$

where $\mathcal{L}_{obj}$ is the cross-entropy loss for object classification, $\mathcal{L}_{verb}$ is the multi-label verb loss defined below, $\mathcal{L}_{L1}$ and $\mathcal{L}_{giou}$ are the L_1 and generalized IoU losses for human and object box regression, and $\mathcal{L}_{dn}$ applies the same supervision to the Sem-DN outputs. The coefficients λ_{obj}, λ_{verb}, λ_{L1}, λ_{giou}, and λ_{dn} balance the corresponding terms.

Since multiple verbs can be valid for the same human-object pair, verb prediction is formulated as multi-label classification and optimized with focal loss:

$$\begin{aligned} \mathcal{L}_{verb} = \frac{1}{N_{pos}} \sum_{q=1}^{N_{pos}} \sum_{i=1}^{K} \Big[-\alpha y_{q,i}(1-\hat{p}_{q,i})^{\gamma}\log(\hat{p}_{q,i}) \\ -(1-\alpha)(1-y_{q,i})\hat{p}^{\gamma}_{q,i}\log(1-\hat{p}_{q,i}) \Big], \end{aligned} \tag{26}$$

where N_{pos} is the number of positive matched instances used for normalization, $y_{q,i} \in \{0,1\}$ and $\hat{p}_{q,i} \in [0,1]$ denote the ground-truth label and predicted probability of verb category i for matched instance q, and α and γ are the balancing and focusing factors.

4. EXPERIMENTS AND ANALYSIS

4.1. Experimental Details

We set the number of layers in both the hybrid encoder and decoder to six, and use 64 object queries for HICO-DET and 100 for V-COCO. The weights of the L_1, generalized IoU, object-classification, and interaction-classification losses are set to 2.5, 1, 1, and 1.5, respectively. The visual backbone is initialized with COCO-pretrained weights, while the remaining parameters are trained from scratch. We optimize SPAF-Net using AdamW with a weight decay of $1e-4$. For HICO-DET, the backbone and the remaining parameters are trained with learning rates of $1e-5$ and $1e-4$, respectively; the learning rate is reduced by a factor of 10 at epochs 40 and 60. For V-COCO, the corresponding learning rates are $5e-6$ and $5e-5$, with one decay at epoch 60. All experiments are conducted on four RTX 3090 GPUs with a batch size of four.

4.2. Comparison with Advanced Methods

We evaluate SPAF-Net on HICO-DET and V-COCO and compare it with representative two-stage and one-stage HOI detectors. The results are reported in Table 1 and Table 2.

Table 1 summarizes the results on HICO-DET. SPAF-Net benefits consistently from stronger visual backbones: the Full mAP under the default setting increases from 38.67% with ResNet-50 to 40.53% with ResNet-101 and 44.73% with Swin-Large-22K. With the strongest backbone, SPAF-Net obtains 44.73% Full mAP and 45.78% Non-Rare mAP in the default setting, together with 47.90% Full mAP and 48.29% Non-Rare mAP in the known-object setting. These results establish the best overall Full and Non-Rare performance among the compared methods. Compared with DiffHOI Yang et al. (2023), SPAF-Net improves the Full mAP by 3.23 and 4.28 percentage points under the default and known-object settings, respectively, while also improving the Non-Rare scores by 3.82 and 4.01 points. Relative to

Table 1
Comparison with advanced methods on the HICO-DET dataset.The letters in the feature column represent: A: Visual Feature, S: Spatial Feature, L: Language Feature, P: Human Posture Feature, M: Multi-scale Feature.The bold and black characters in the table indicate the best test results for each indicator.

Method	Backbone	Feature	Epoch	Default			Known Object		
				Full	Rare	Non-Rare	Full	Rare	Non-Rare
Two-Stage Methods									
CATN Dong, Li, Xu, Zhang, Yan, Zhong and Zou (2022)	ResNet-50	A	12	31.86	25.15	33.84	34.44	27.69	36.45
STIP Zhang, Pan, Yao, Huang, Mei and Chen (2022c)	ResNet-50	A+L+S	30	32.22	28.15	33.43	35.29	31.43	36.45
DEFR Jin, Chen, Wang, Wang, Yu, Liang, Hwang and Liu (2022)	ViT-B/16	A+L	20	32.35	33.45	32.02	-	-	-
UPT Zhang, Campbell and Gould (2022a)	ResNet-101	A	20	32.62	28.62	33.81	36.08	31.41	37.47
VIPLO Park et al. (2023)	ViT-B/16	A+P	15	37.22	35.45	37.75	40.61	38.82	41.15
RmLR Cao, Tang, Yang, Su, You, Lu and Xu (2023)	ResNet-101	A+L	20	37.41	28.81	39.97	38.69	31.27	40.91
ADA-CM Lei et al. (2023)	ViT-L	A+L	15	38.40	37.52	38.66	-	-	-
SCTC++ Ren et al. (2024)	ViT-B/16	A+L	30	40.32	37.15	41.27	43.18	39.90	44.33
One-Stage Methods									
QPIC Tamura et al. (2021)	ResNet-101	A	150	29.90	23.92	31.69	32.58	26.06	34.27
MSTR Kim, Mun, On, Shin, Lee and Kim (2022)	ResNet-50	A+M	50	31.17	25.31	32.92	34.02	28.83	35.57
RLIP-ParSe Yuan, Zhang, Wang, Albanie, Pan, Feng, Jiang, Ni, Zhang and Zhao (2023)	ResNet-50	A+L	90	32.84	34.63	26.85	-	-	-
MUREN Kim, Jung and Cho (2023)	ResNet-50	A	100	32.87	28.67	34.12	35.52	30.88	36.91
TED-Net Wang, Liu and Lei (2024)	ResNet-50	A+L	120	34.00	29.88	35.24	37.13	33.63	38.18
UAHOI Chen, Chen and Yang (2024)	ResNet-50	A	-	34.19	31.54	35.27	37.44	34.18	38.65
HOICLIP Ning et al. (2023)	ResNet-50	A+L	90	34.69	31.12	35.74	37.61	34.47	38.54
GEN-VLKT-L Liao et al. (2022)	ResNet-101	A+L	90	34.95	31.18	36.08	38.22	34.36	39.37
CLIP4HOI Mao, Deng, Zhou, Li, Fang and Li (2023)	ResNet-50	A+L	100	35.33	33.95	35.74	37.19	35.27	37.77
LactFormer Peng, Chen, Tang, Yu, Wang, Xie and Fang (2025)	ResNet-101	A+L	20	35.62	33.90	36.13	39.30	36.37	40.16
QAHQI Junwen and Keiji (2023)	Swin-Large-22K	A+M	150	35.78	29.80	37.56	37.59	31.36	39.36
ERNet Lim, Baskaran, Lim, Wong, See and Tistarelli (2023)	EfficientNetV2-XL	A+M	100	35.92	30.12	38.29	-	-	-
FGAHOI Ma et al. (2023)	Swin-Large-22K	A+S+M	190	37.18	30.71	39.11	38.93	31.93	41.02
DiffHOI Yang, Li, Yang, Zeng, Zhang and Zhang (2023)	Swin-Large-22K	A+L	-	41.50	39.96	41.96	43.62	41.41	44.28
PVIC Zhang et al. (2023)	Swin-Large-22K	A+S+M	30	44.32	**44.61**	44.24	47.81	**48.38**	47.64
SPAF-Net	ResNet50	A+M+L	80	38.67	34.83	39.84	41.07	37.13	41.95
SPAF-Net	ResNet101	A+M+L	80	40.53	36.74	41.70	41.87	38.75	42.89
SPAF-Net	Swin-Large-22K	A+M+L	120	**44.73**	41.32	**45.78**	**47.90**	42.39	**48.29**

Table 2
Comparison with advanced methods on the V-COCO dataset. Bold values denote the best results for each evaluation metric.

Method	Backbone	Epoch	AP^{S1}_{role}	AP^{S2}_{role}
Two-Stage Methods				
ADA-CM Lei et al. (2023)	ViT-L	15	58.57	63.97
UPT Zhang et al. (2022a)	ResNet-101	20	61.3	67.1
OCN Yuan, Wang, Ni and Xu (2022b)	ResNet-101	80	65.3	67.1
ViPLO Park et al. (2023)	ViT-B/16	15	62.2	68.0
One-Stage Methods				
QAHOI Junwen and Keiji (2023)	ResNet-50	150	58.2	58.7
FGAHOI Ma et al. (2023)	Swin-Large	190	60.5	61.2
RLIP-ParSe Yuan et al. (2023)	ResNet-50	90	61.9	64.2
MSTR Kim et al. (2022)	ResNet50	50	62.0	65.2
UAHOI Chen et al. (2024)	ResNet50	-	62.6	66.7
TED-Net Wang et al. (2024)	ResNet50	120	63.41	64.96
HOICLIP Ning et al. (2023)	ResNet-50	90	63.5	64.8
GEN-VLKT-L Liao et al. (2022)	ResNet101	90	63.6	65.9
SOV-STG Chen et al. (2025)	ResNet50	30	63.8	65.7
PVIC Zhang et al. (2023)	Swin-Large-22k	30	64.1	70.2
RLIPv2-ParSeDA Yuan et al. (2023)	Swin-Large	20	64.4	66.5
DiffHOI Yang et al. (2023)	Swin-Large-22K	-	65.7	68.2
SPAF-Net	ResNet50	80	63.9	67.3
SPAF-Net	ResNet101	80	64.3	68.3
SPAF-Net	Swin-Large-22K	100	**66.9**	**71.3**

Table 3
The effectiveness of the method or module is evaluated using the HICO-DET test set.

Baseline	TDE-GF	SD-HNM	SG-ACF	Default		
				Full	Rare	Non-Rare
√	-	-	-	32.17	28.28	34.06
√	√	-	-	35.68	30.45	36.75
√	√	√	-	36.17	32.15	37.42
√	√	√	√	38.67	34.83	39.84

Table 4
The effectiveness of the method or module is evaluated using the V-COCO test set.

Baseline	TDE-GF	SD-HNM	SG-ACF	AP^{S1}_{role}	AP^{S2}_{role}
√	-	-	-	61.9	64.7
√	√	-	-	63.7	67.2
√	√	√	-	65.1	69.3
√	√	√	√	66.9	71.3

SCTC++ Ren et al. (2024), the gains in Full mAP are 4.41 and 4.72 points. SPAF-Net does not achieve the highest Rare score, where PVIC Zhang et al. (2023) remains stronger, this indicates that rare-category recognition still leaves room for improvement despite the overall gains.

The comparison also highlights the effect of the training and representation design. Although PVIC obtains a higher Rare mAP, SPAF-Net achieves a stronger balance between rare and non-rare categories without relying on additional synthetic training data. In particular, its non-rare performance exceeds PVIC by 1.54 points in the default setting and 0.65 points in the known-object setting, while its Full scores are comparable or better.

Table 2 reports the results on V-COCO. With Swin-Large-22K, SPAF-Net achieves 66.9% AP^{S1}_{role} and 71.3% AP^{S2}_{role}, surpassing DiffHOI Yang et al. (2023) by 1.2 and 3.1 points, respectively, and PVIC Zhang et al. (2023) by 2.8 and 1.1 points. The gains over the image-only MSTR Kim et al.

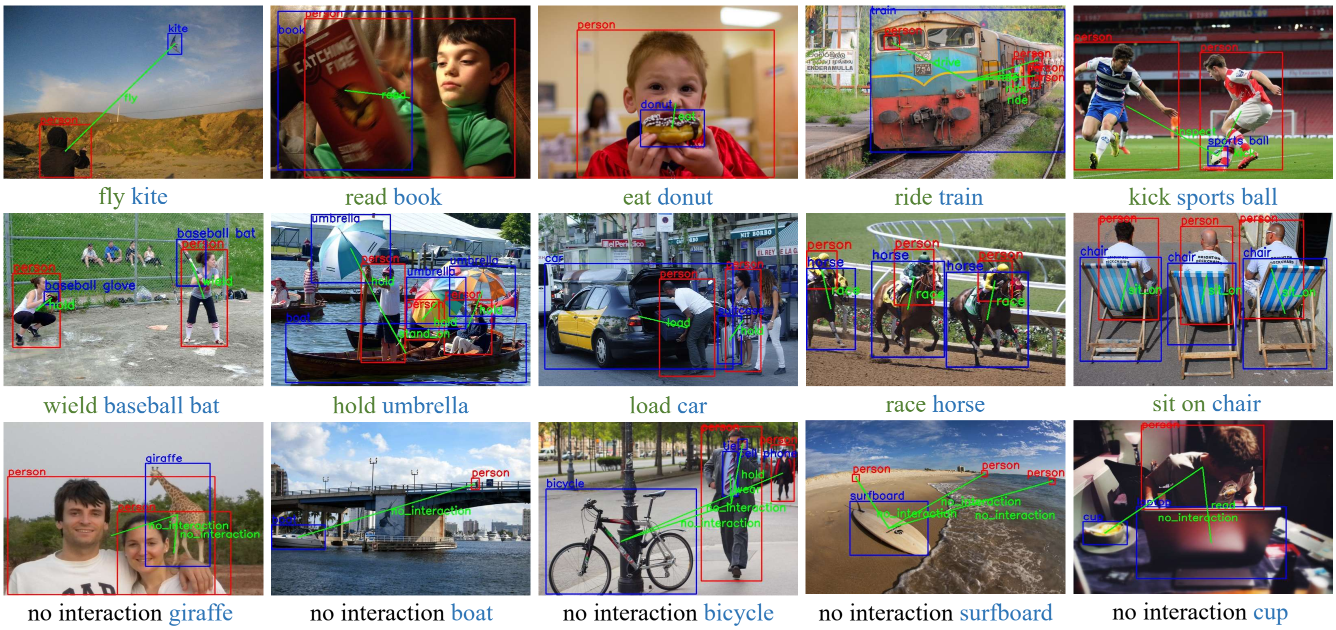

Figure 6: The visualization of predicted results of **HICO-DET**. The first row shows visualizations of one or more subjects interacting with the same object. The second row presents multiple interactions occurring among multiple individuals and multiple objects. The third row demonstrates the visualization outcomes incorporating the interaction category of "no interaction.

(2022) are 4.9 and 6.1 points. The consistent improvements on both role evaluation settings suggest that the proposed textual prior injection and spatially guided context retrieval improve not only interaction classification but also human-object role assignment.

4.3. Ablation Experiment

We conduct a cumulative ablation study by adding the three proposed components one at a time. For efficiency, all ablations use ResNet-50 and retain the same training protocol. The baseline is the visual detector without text-enhanced fusion, semantic denoising, and spatially guided context fusion. Results are reported in Tables 3 and 4.

(1) Text-Enhanced Dual-Stream Encoder with Gated Fusion: Adding TDE-GF improves the HICO-DET Full, Rare, and Non-Rare mAP from 32.17%, 28.28%, and 34.06% to 35.68%, 30.45%, and 36.75%, corresponding to gains of 3.51, 2.17, and 2.69 points. On V-COCO, AP^{S1}_{role} and AP^{S2}_{role} increase by 1.8 and 2.5 points. These improvements show that gated residual fusion can exploit textual priors while retaining the visual evidence required for fine-grained interaction recognition.

(2) Semantic Denoising for Hard-Negative Mining: Introducing SD-HNM on top of TDE-GF further raises the HICO-DET scores to 36.17%/32.15%/37.42% for the Full/Rare/Non-Rare splits. The Rare score increases by 1.70 points, which is the largest gain among the three HICO-DET splits, indicating that semantically similar label perturbations provide informative hard negatives for long-tailed categories. The corresponding V-COCO scores improve from 63.7%/67.2% to 65.1%/69.3%.

(3) Spatially Guided Adaptive Context Fusion: Finally, SG-ACF improves the complete model to 38.67% Full, 34.83% Rare, and 39.84% Non-Rare mAP on HICO-DET, yielding additional gains of 2.50, 2.68, and 2.42 points over the preceding variant. On V-COCO, it contributes 1.8 and 2.0 points in AP^{S1}_{role} and AP^{S2}_{role}, respectively. The cumulative gains over the baseline reach 6.50, 6.55, and 5.78 points on HICO-DET and 5.0 and 6.6 points on V-COCO. This result verifies that region-aware VLM retrieval and adaptive branch routing effectively suppress background interference and provide complementary context to the human, object, and interaction branches.

4.4. Qualitative Analysis

Figure 6 and Figure 7 present the final prediction results of SPAF-Net on the HICO-DET and V-COCO datasets, while Figure 8 depicts the cross-attention maps for the three decoder branches and the SG-ACF module.

In Figure 6, the first row demonstrates that SPAF-Net can reliably detect HOIs involving either single or multiple subjects interacting with the same object, even under challenging conditions such as long-range engagements and small-scale targets (e.g., fly a kite). The fourth and fifth examples further emphasize the framework's capacity to differentiate visually similar interaction categories (e.g., blocking vs. kicking, and driving vs. riding), confirming its fine-grained action discrimination and robustness to intra-class visual similarity. The second row depicts complex scenes with multiple subjects and objects, where SPAF-Net successfully recognizes all valid HOI pairs, reflecting its resilience and adaptability in highly interactive environments. The third row highlights "no-interaction" scenarios, in which the third

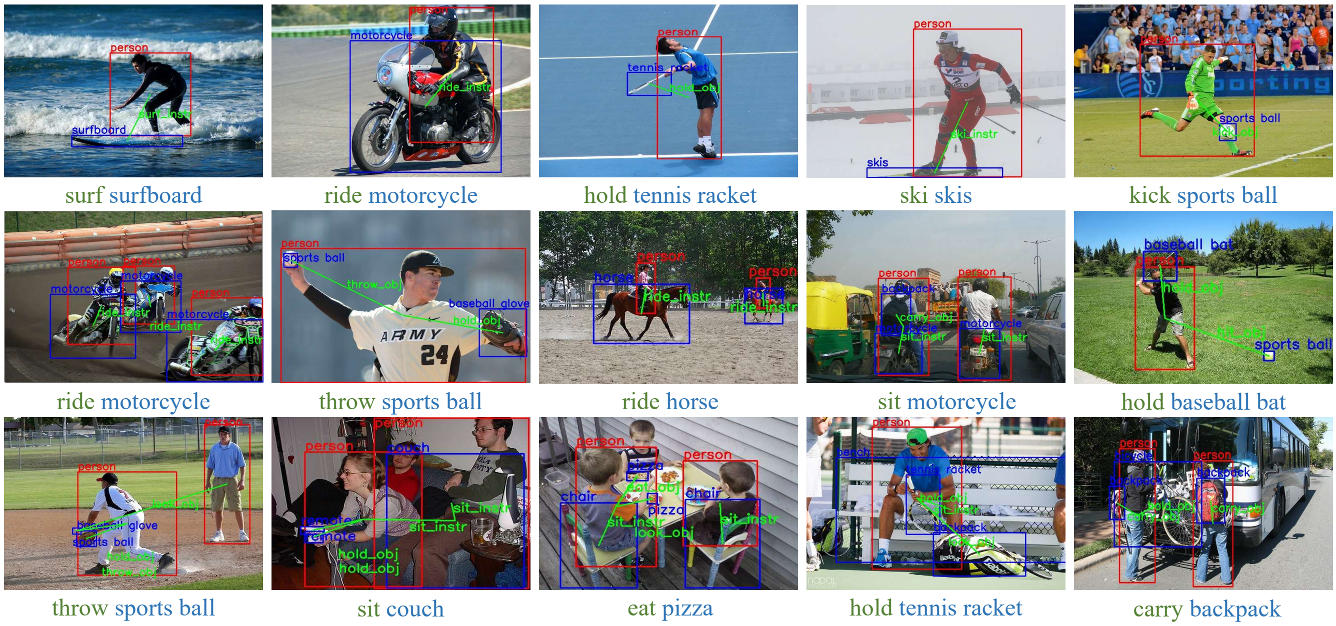

Figure 7: The visualization of predicted results of **V-COCO**. The first row illustrates the visualization of a single interaction scenario. The second and third row represent multiple interactions occurring among multiple individuals and multiple objects.

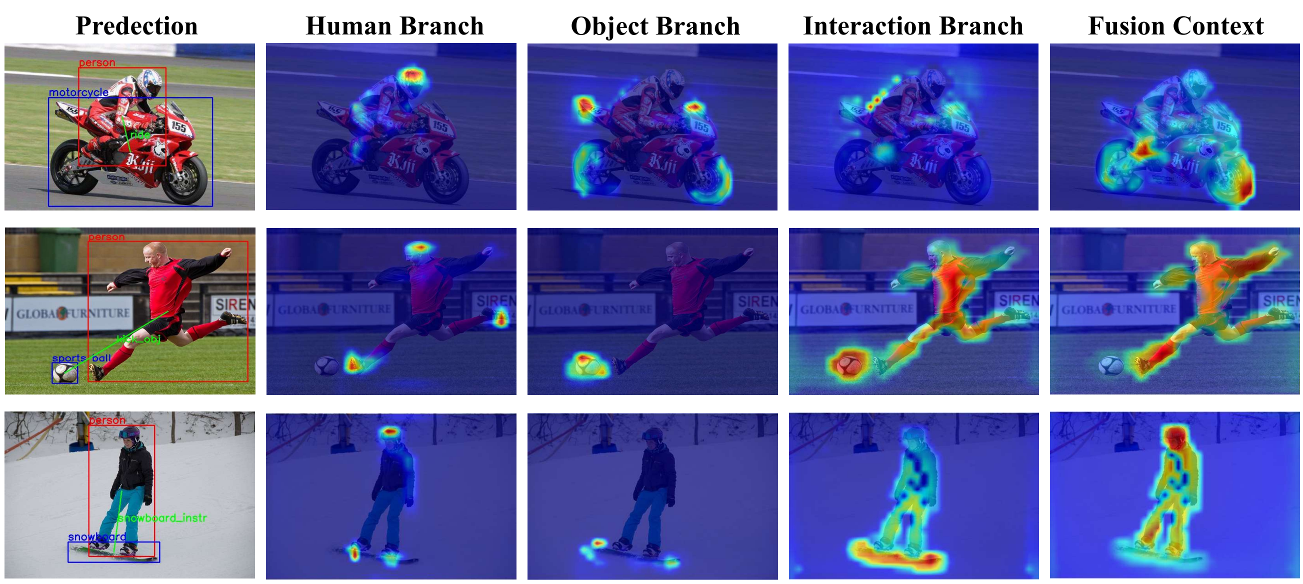

Figure 8: Visualization of the cross-attention weights for each branch and the SG-ACF contextual information. In the figure, "Prediction" depicts the predicted results. The Human, Object and Interaction Branch represent the cross-attention weight visualizations for the human, object, and interaction branches, respectively. "Fusion Context" illustrates the cross-attention weight visualization for the SG-ACF contextual information.

and fifth samples correctly identify the absence of interaction between certain subject–object pairs (e.g., bicycle or cup), attesting to the model's reliability in negative-sample detection.

Figure 7 displays prediction results for the V-COCO dataset, where the first row contains single-interaction cases, while the second and third rows present multi-interaction examples with greater visual complexity. Combined, these results substantiate SPAF-Net's capacity for accurate HOI detection, fine-grained semantic reasoning, and generalization across different interaction scales and complexities.

The cross-attention maps of each decoder branch and the SG-ACF module are visualized in Figure 8. The Human and Object Branch attention maps show that both branches effectively focus on their respective instances, facilitating their accurate localization and categorization. In contrast, the Interaction Branch attention maps concentrate on the interaction regions between humans and objects, enabling precise identification of specific actions. The Fusion Context visualization from the SG-ACF module prominently highlights the entire region of HOI instances, indicating its capacity to capture global contextual information for HOI detection. This context-aware integration augments each branch with supplementary semantic cues, enhancing predictive accuracy and ultimately yielding more precise HOI detection results.

5. Conclusion

HOI detection enables machines to move beyond object localization toward a deeper understanding of dynamic visual scenes. To address the semantic ambiguity, background interference, and long-tailed distribution challenges in this task, we propose SPAF-Net, an end-to-end framework that combines text-enhanced dual-stream encoding, semantic denoising, and spatially guided adaptive context fusion. The gated cross-modal encoder injects textual priors while preserving fine-grained visual evidence, SD-HNM constructs informative hard negatives for long-tailed interaction categories, and SG-ACF retrieves region-specific VLM context and routes it to the human, object, and interaction branches. Extensive experiments on HICO-DET and V-COCO demonstrate the effectiveness of SPAF-Net and its superior overall performance.

SPAF-Net's design facilitates its extension to diverse applications like human-robot collaboration and human behavior analysis. In future work, we will explore integrating prior knowledge from contrastive language–image pre-training to enhance the zero-shot generalization capability of SPAF-Net, thereby further improving HOI detection performance.

CRediT authorship contribution statement

Ke Wang: Writing – review & editing, Project administration, Resources, Funding acquisition. **Yuhao Jiang:** Writing – review & editing, Writing – original draft, Methodology, Formal analysis. **Chuang Qiu:** Writing – review & editing, Validation. **Weilin Gao:** Writing – review & editing, Methodology, Validation.

References

Cao, Y., Tang, Q., Yang, F., Su, X., You, S., Lu, X., Xu, C., 2023. Re-mine, learn and reason: Exploring the cross-modal semantic correlations for language-guided hoi detection, in: Proceedings of the IEEE/CVF international conference on computer vision, pp. 23492–23503.

Chao, Y.W., Liu, Y., Liu, X., Zeng, H., Deng, J., 2018. Learning to detect human-object interactions, in: 2018 ieee winter conference on applications of computer vision (wacv), IEEE. pp. 381–389.

Chen, J., Wang, Y., Yanai, K., 2025. Focusing on what to decode and what to train: Sov decoding with specific target guided denoising and vision language advisor, in: 2025 IEEE/CVF Winter Conference on Applications of Computer Vision (WACV), IEEE. pp. 9416–9425.

Chen, M., Chen, M., Yang, Y., 2024. Uahoi: Uncertainty-aware robust interaction learning for hoi detection. Computer Vision and Image Understanding 247, 104091.

Dong, L., Li, Z., Xu, K., Zhang, Z., Yan, L., Zhong, S., Zou, X., 2022. Category-aware transformer network for better human-object interaction detection, in: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition, pp. 19538–19547.

Guo, M., Lan, H., Yang, C., Liu, J., Gao, F., 2022. As-net: fast photoacoustic reconstruction with multi-feature fusion from sparse data. IEEE Transactions on Computational Imaging 8, 215–223.

Gupta, S., Malik, J., 2015. Visual semantic role labeling. arXiv preprint arXiv:1505.04474 .

Hou, Z., Peng, X., Qiao, Y., Tao, D., 2020. Visual compositional learning for human-object interaction detection, in: European Conference on Computer Vision, Springer. pp. 584–600.

Hou, Z., Yu, B., Qiao, Y., Peng, X., Tao, D., 2021. Affordance transfer learning for human-object interaction detection, in: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition, pp. 495–504.

Jin, Y., Chen, Y., Wang, L., Wang, J., Yu, P., Liang, L., Hwang, J.N., Liu, Z., 2022. The overlooked classifier in human-object interaction recognition. arXiv preprint arXiv:2203.05676 .

Junwen, C., Keiji, Y., 2023. Qahoi: Query-based anchors for human-object interaction detection. IEICE Proceedings Series 78.

Kim, B., Lee, J., Kang, J., Kim, E.S., Kim, H.J., 2021. Hotr: End-to-end human-object interaction detection with transformers, in: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition, pp. 74–83.

Kim, B., Mun, J., On, K.W., Shin, M., Lee, J., Kim, E.S., 2022. Mstr: Multi-scale transformer for end-to-end human-object interaction detection, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 19578–19587.

Kim, S., Jung, D., Cho, M., 2023. Relational context learning for human-object interaction detection, in: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition, pp. 2925–2934.

Lei, T., Caba, F., Chen, Q., Jin, H., Peng, Y., Liu, Y., 2023. Efficient adaptive human-object interaction detection with concept-guided memory, in: Proceedings of the IEEE/CVF international conference on computer vision, pp. 6480–6490.

Liao, Y., Zhang, A., Lu, M., Wang, Y., Li, X., Liu, S., 2022. Gen-vlkt: Simplify association and enhance interaction understanding for hoi detection, in: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition, pp. 20123–20132.

Lim, J., Baskaran, V.M., Lim, J.M.Y., Wong, K., See, J., Tistarelli, M., 2023. Ernet: An efficient and reliable human-object interaction detection network. IEEE Transactions on Image Processing 32, 964–979.

Ma, S., Wang, Y., Wang, S., Wei, Y., 2023. Fgahoi: Fine-grained anchors for human-object interaction detection. IEEE Transactions on Pattern Analysis and Machine Intelligence 46, 2415–2429.

Mao, Y., Deng, J., Zhou, W., Li, L., Fang, Y., Li, H., 2023. Clip4hoi: Towards adapting clip for practical zero-shot hoi detection. Advances in Neural Information Processing Systems 36, 45895–45906.

Ning, S., Qiu, L., Liu, Y., He, X., 2023. Hoiclip: Efficient knowledge transfer for hoi detection with vision-language models, in: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition, pp. 23507–23517.

Park, J., Park, J.W., Lee, J.S., 2023. Viplo: Vision transformer based pose-conditioned self-loop graph for human-object interaction detection, in: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition, pp. 17152–17162.

Peng, W., Chen, Q., Tang, K., Yu, Z., Wang, Y., Xie, Y., Fang, M., 2025. Lactformer: Language-aware context learning with multi-cue transformers for hoi detection. IEEE Transactions on Consumer Electronics 71, 3617–3630. doi:10.1109/TCE.2025.3571778.

Radford, A., Kim, J.W., Hallacy, C., Ramesh, A., Goh, G., Agarwal, S., Sastry, G., Askell, A., Mishkin, P., Clark, J., et al., 2021. Learning transferable visual models from natural language supervision, in: International conference on machine learning, PmLR. pp. 8748–8763.

Ren, W., Luo, J., Jiang, W., Qu, L., Han, Z., Tian, J., Liu, H., 2024. Learning self-and cross-triplet context clues for human-object interaction detection. IEEE Transactions on Circuits and Systems for Video Technology 34, 9760–9773.

Su, Z., Wang, Y., Xie, Q., Yu, R., 2022. Pose graph parsing network for human-object interaction detection. Neurocomputing 476, 53–62.

Tamura, M., Ohashi, H., Yoshinaga, T., 2021. Qpic: Query-based pairwise human-object interaction detection with image-wide contextual information, in: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition, pp. 10410–10419.

Ulutan, O., Iftekhar, A., Manjunath, B.S., 2020. Vsgnet: Spatial attention network for detecting human object interactions using graph convolutions, in: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition, pp. 13617–13626.

Wang, H., Jiao, L., Liu, F., Li, L., Liu, X., Ji, D., Gan, W., 2021. Ipgn: Interactiveness proposal graph network for human-object interaction detection. IEEE Transactions on Image Processing 30, 6583–6593.

Wang, Y., Liu, Q., Lei, Y., 2024. Ted-net: Dispersal attention for perceiving interaction region in indirectly-contact hoi detection. IEEE Transactions on Circuits and Systems for Video Technology 34, 5603–5615.

Yang, J., Li, B., Yang, F., Zeng, A., Zhang, L., Zhang, R., 2023. Boosting human-object interaction detection with text-to-image diffusion model. arXiv preprint arXiv:2305.12252 .

Yuan, H., Jiang, J., Albanie, S., Feng, T., Huang, Z., Ni, D., Tang, M., 2022a. Rlip: Relational language-image pre-training for human-object interaction detection. Advances in Neural Information Processing Systems 35, 37416–37431.

Yuan, H., Wang, M., Ni, D., Xu, L., 2022b. Detecting human-object interactions with object-guided cross-modal calibrated semantics, in: Proceedings of the AAAI Conference on artificial intelligence, pp. 3206–3214.

Yuan, H., Zhang, S., Wang, X., Albanie, S., Pan, Y., Feng, T., Jiang, J., Ni, D., Zhang, Y., Zhao, D., 2023. Rlipv2: Fast scaling of relational language-image pre-training, in: Proceedings of the IEEE/CVF international conference on computer vision, pp. 21649–21661.

Zhang, A., Liao, Y., Liu, S., Lu, M., Wang, Y., Gao, C., Li, X., 2021. Mining the benefits of two-stage and one-stage hoi detection. Advances in neural information processing systems 34, 17209–17220.

Zhang, F.Z., Campbell, D., Gould, S., 2022a. Efficient two-stage detection of human-object interactions with a novel unary-pairwise transformer, in: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition, pp. 20104–20112.

Zhang, F.Z., Yuan, Y., Campbell, D., Zhong, Z., Gould, S., 2023. Exploring predicate visual context in detecting of human-object interactions, in: Proceedings of the IEEE/CVF international conference on computer vision, pp. 10411–10421.

Zhang, H., Li, F., Liu, S., Zhang, L., Su, H., Zhu, J., Ni, L.M., Shum, H.Y., 2022b. Dino: Detr with improved denoising anchor boxes for end-to-end object detection. arXiv preprint arXiv:2203.03605 .

Zhang, Y., Pan, Y., Yao, T., Huang, R., Mei, T., Chen, C.W., 2022c. Exploring structure-aware transformer over interaction proposals for human-object interaction detection, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 19548–19557.

Zhao, Y., Lv, W., Xu, S., Wei, J., Wang, G., Dang, Q., Liu, Y., Chen, J., 2024. Detrs beat yolos on real-time object detection, in: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition, pp. 16965–16974.

Zou, C., Wang, B., Hu, Y., Liu, J., Wu, Q., Zhao, Y., Li, B., Zhang, C., Zhang, C., Wei, Y., et al., 2021. End-to-end human object interaction detection with hoi transformer, in: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition, pp. 11825–11834.